Navaneeth S

Data Scientist | AI/ML Engineer

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LinkedIn | GitHub | Portfolio

Summary

AI/ML engineer with published deep learning research in SAR-based flood segmentation (submitted to IEEE Geoscience and Remote Sensing Letters) and hands-on experience building production-grade RAG and AI-driven forecasting systems at GE Aerospace. Strong full-stack ML foundation spanning model design, deployment, and cloud infrastructure, with a track record as a finalist in AI hackathons. Passionate about applying AI to solve real-world enterprise and research problems.

Research Publication

Urban Flood Detection Using SAR Imagery with Multi-Scale Feature Extraction and Dual-Encoder U-Net

Navaneeth S, Dr. Akshara P Byju Submitted to IEEE Geoscience and Remote Sensing Letters (2026)

- Designed a novel MSVDP (Multi-Scale SVD Profile) dual-encoder cross-attention U-Net that fuses SAR radiometric backscatter and SLIC-superpixel structural features for 3-class urban flood segmentation.
- Achieved 0.6486 mIoU and 98.77% pixel accuracy on the UrbanSARFloods benchmark, improving mIoU by 23.94% over a Simple U-Net baseline and 6.68% over an SVD-only model. Obtained the highest F1-score (0.737) among nine compared architectures, including U-Net++, MANet, FPN, PSPNet, and DeepLabv3+.
- Developed the complete pipeline in PyTorch, including SLIC multi-scale feature extraction, dual-stream residual encoders, cross-attention fusion, and a composite Focal + Dice + Tversky + Cross-Entropy loss for severe class imbalance. Trained using mixed precision and EMA on an NVIDIA T4 GPU.

Experience

Data Science Intern - GE Aerospace Jan 2026 – Present

- Designed and deployed a production-grade Retrieval-Augmented Generation (RAG) assistant (LangChain, LangGraph, Amazon Bedrock, PostgreSQL/pgvector) for company-wide travel and policy query resolution, owning the pipeline end-to-end from data preprocessing through deployment on AWS Lambda and S3.
- Reduced support workload by automating employee query resolution, lowering dependence on manual ticket handling and helping maintain support volume below 1,000 cases/month.
- Contributing to an AI-enabled, driver-based forecasting platform for Corporate Functions Purchased Services, automating accrual, prepaid, and PO-level forecasting workflows in collaboration with Corporate FP&A teams.
- Supporting a transformation initiative targeting a 50% reduction in approximately 5,500 annual hours spent on manual forecasting, reconciliation, and transaction-monitoring, via data pipeline development and forecasting model design.

Projects

SmartFin – AI-Powered Personal Finance Assistant

- Built an LSTM-based time-series forecasting model to predict personal spending patterns, trained and validated on real transaction data.
- Developed a full-stack solution using Python, Flask, and MongoDB for personalized financial planning and forecasting.
- Designed dynamic analytics dashboards surfacing expenditure forecasts and category-wise spending insights.

SignAI – Real-Time Sign Language Interpreter

- Engineered a real-time sign language interpretation system using OpenCV, MediaPipe, and Flask for gesture recognition and translation.
- Achieved inference latency below 100 ms for live gesture-to-text and gesture-to-speech conversion.
- Presented a working prototype as a finalist at the Tredence Infinity AI Hackathon 2025.

Education

- **M.Tech in CSE with AI and ML** — 2024 – 2026
  Amrita School of Computing, Amritapuri — Amrita University, 8.91 CGPA
- **B.Tech in Computer Science and Engineering** — 2020 – 2024
  College of Engineering, Kallooppara — KTU, 7.68 CGPA
- **Senior Secondary Education** — 2020
  Placid Vidya Vihar, Changanassery — CBSE, 80.2%
- **Secondary Education** — 2018
  SBHSS, Mannar — CBSE, 92.2%

Technical Skills

- **Programming Languages:** Java, Python
- **AI/ML & Data:** PyTorch, TensorFlow, Flask, Scikit-learn, Computer Vision (OpenCV, MediaPipe, YOLOv5), Agentic AI, RAG, LangChain, LangGraph, Pandas, NumPy
- **Cloud & Databases:** AWS (S3, Lambda, Bedrock), GCP, SQL, PostgreSQL (pgvector), MongoDB, Firebase Firestore
- **Tools & Platforms:** Git, GitHub

Soft Skills

Leadership, Adaptability, Time Management, Self Assessment

Achievements & Certifications

- Finalist of **Kochi Hackathon '25**
  Developed Voxel Web Reader, an AI-powered Chrome extension using YOLOv5-based image description for accessible web reading.
- **Build with AI Bootcamp** – Google Developers
  Demonstrated hands-on experience in building AI agents, architecting workflows, and integrating Generative AI into production-ready systems.
- Introduction to Large Language Models — NPTEL (IIT), Online Certification.

Languages

Malayalam (Native Proficiency), **English** (Full Professional Proficiency), **Hindi** (Working Proficiency)

Interests

Travelling, Philately, Numismatics, Exploring AI/ML innovations and emerging technologies